# A. Sizing code verification: % errors

## 1. Design point output aircraft verification % errors (using book takeoff gross weight)

For the table below, the takeoff gross weight stays fixed at the true value for each aircraft. The reason the empty weight error is a bit high for the CJ-3 is because the empty weight for subsonic transports were calibrated by applying the Raymer empty weight component fudge factors using the Boeing 737-200 as a baseline. The empty weight for the SR-71 was estimated using the XB-70 as a baseline. The empty weight for the TU-144D was estimated using the Concorde as a baseline. According to research, a high error for the empty weight for the B-58 is expected.

- TOGW = Takeoff gross weight
- $W_e$ = Empty weight
- $W_f$ = Fuel weight
- Alt = design altitude
- R = range
- L/D = Lift to drag ratio
- TSFC = Thrust specific fuel consumption

*Table 1: Percent errors of predicted performance parameters relative to reference aircraft data (using true takeoff gross weight).*

| *Aircraft* | *TOGW* | $W_e$ | $W_f$ | *Alt* | *R* | *L/D* | *TSFC* |
|---|---|---|---|---|---|---|---|
| *Supersonic fighters* | | | | | | | |
| *XB-70* | Book | 2.6 | 4.4 | Book | Book | 5.1 | 4.5 |
| *SR-71* | Book | 0.6 | 4.3 | Book | Book | 2.3 | 2.8 |
| *Supersonic transports* | | | | | | | |
| *Concorde* | Book | 0.0 | 5.5 | Book | Book | 0.3 | 0.8 |
| *TU-144D* | Book | 5.0 | 4.6 | Book | Book | 2.5 | 2.7 |
| *Subsonic transports* | | | | | | | |
| *B737-200* | Book | 2.3 | 2.4 | Book | Book | 1.7 | 2.9 |
| *B777-200er* | Book | 7.2 | 7.6 | Book | Book | 0.8 | 2.3 |
| *2000 EX* | Book | 6.9 | 2.7 | Book | Book | 0.3 | 0.3 |
| *Subsonic general aviation* | | | | | | | |
| *CJ3* | Book | 9.2 | 2.6 | Book | Book | 0.4 | 6.7 |

## 2. Geometric wing output verification % errors (using book takeoff gross weight)

The reason for the high error for the root chord length of the Concorde in the table below is because the Concorde has an ogive wing while the formulas in the sizing code are for a trapezoidal and triangular delta wing.

- S = wing area
- $b$ = wingspan
- $c_r$ = root chord length
- $\Lambda$ = leading edge sweep angle

*Table 2: Percent errors of predicted wing geometric variables relative to reference aircraft data (using true takeoff gross weight).*

| *Aircraft* | *S* | *b* | $c_r$ | $\Lambda$ |
|---|---|---|---|---|
| | | *Supersonic fighters* | | |
| *XB-70* | 2.2 | 4.3 | 1.9 | - |
| *SR-71* | 3.4 | 1.2 | 5.1 | - |
| | | *Supersonic transports* | | |
| *Concorde* | 0.7 | 0.2 | 12.3 | — |
| *TU-144* | 0.3 | 0.0 | 1.7 | — |
| | | *Subsonic transports* | | |
| *B737-200* | 1.2 | 0.6 | 4.5 | 0.0 |
| *B777-200er* | 0.8 | 0.4 | 0.4 | 0.1 |
| *2000 EX* | 0.4 | 0.2 | 0.2 | — |
| | | *Subsonic general aviation* | | |
| *CJ-3* | 0.5 | 0.3 | 0.3 | — |

## 3. Design point output aircraft verification % errors (takeoff gross weight free to iterate)

For the table below, the takeoff gross weight is free to iterate which significantly multiplies the magnitude of the percent errors from the tables above due to the snowball effect.

- TOGW = Takeoff gross weight
- $W_e$ = Empty weight
- $W_f$ = Fuel weight
- Alt = design altitude
- R = range
- L/D = Lift to drag ratio
- TSFC = Thrust specific fuel consumption

*Table 3: Percent errors of predicted performance parameters relative to reference aircraft data (allowing takeoff gross weight to iterate).*

| *Aircraft* | *TOGW* | $W_e$ | $W_f$ | *Alt* | *R* | *L/D* | *TSFC* |
|---|---|---|---|---|---|---|---|
| | | | *Supersonic fighters* | | | | |
| *XB-70* | 0.9 | 2.2 | 3.8 | 5.6 | 1.0 | 4.7 | 2.7 |
| *SR-71* | 3.0 | 0.5 | 4.8 | 7.3 | 1.5 | 2.7 | 3.4 |
| | | | *Supersonic transports* | | | | |
| *Concorde* | 4.9 | 2.5 | 7.6 | 1.3 | 3.2 | 2.8 | 1.4 |
| *TU-144D* | 16.0 | 13.1 | 12.3 | 4.8 | 1.7 | 10.6 | 9.5 |
| | | | *Subsonic transports* | | | | |
| *B737-200* | 4.9 | 4.6 | 7.0 | 0.9 | 3.3 | 4.1 | 3.3 |
| *B777-200er* | 17.3 | 16.7 | 17.7 | 13.9 | 3.9 | 9.8 | 4.7 |
| *2000 EX* | 9.3 | 10.5 | 8.1 | 0.9 | 4.3 | 5.8 | 1.6 |
| | | | *Subsonic general aviation* | | | | |
| *CJ3* | 19.4 | 17.2 | 25.0 | 19.6 | 2.5 | 2.2 | 5.0 |

## 4. Geometric wing output verification % errors (takeoff gross weight free to iterate)

For the table below, the takeoff gross weight is free to iterate which significantly multiplies the magnitude of the percent errors from the tables above due to the snowball effect. The reason for the high error for the root chord length of the Concorde in the table below is because the Concorde has an ogive wing while the formulas in the sizing code are for a trapezoidal and triangular delta wing.

- S = wing area
- $b$ = wingspan
- $c_r$ = root chord length
- $\Lambda$ = leading edge sweep angle

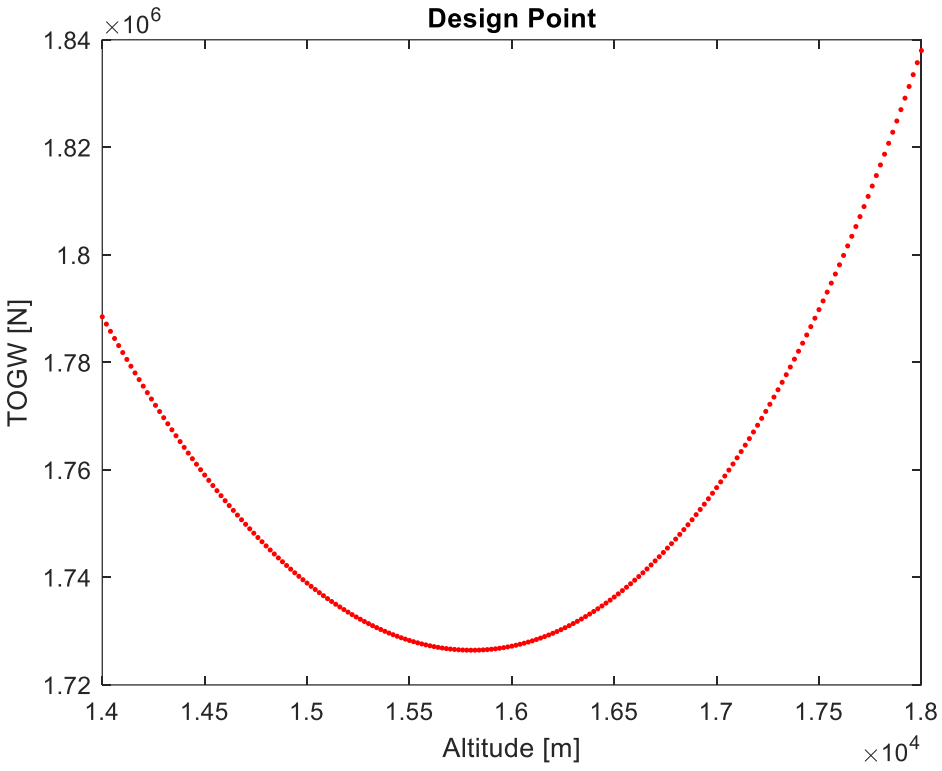
*Table 4: Percent errors of predicted wing geometric variables relative to reference aircraft data (allowing takeoff gross weight to iterate).*

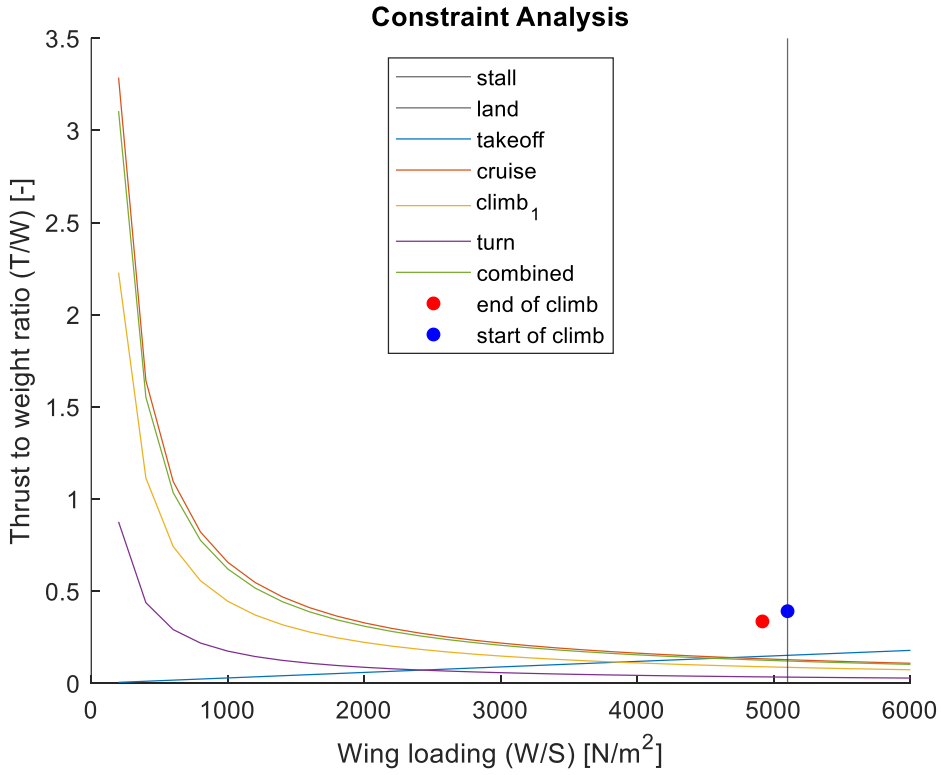
| *Aircraft* | *S* | *b* | $c_r$ | $\Lambda$ |
|---|---|---|---|---|
| | | *Supersonic fighters* | | |
| *XB-70* | 3.1 | 4.8 | 1.5 | — |
| *SR-71* | 0.3 | 0.3 | 3.5 | — |
| | | *Supersonic transports* | | |
| *Concorde* | 5.5 | 2.6 | 14.5 | — |
| *TU-144* | 16.2 | 8.4 | 9.9 | — |
| | | *Subsonic transports* | | |
| *B737-200* | 6.0 | 3.1 | 6.9 | 0.0 |
| *B777-200er* | 16.6 | 8.7 | 8.7 | 0.1 |
| *2000 EX* | 9.0 | 4.6 | 4.6 | — |
| | | *Subsonic general aviation* | | |
| *CJ-3* | 19.9 | 10.5 | 10.5 | — |

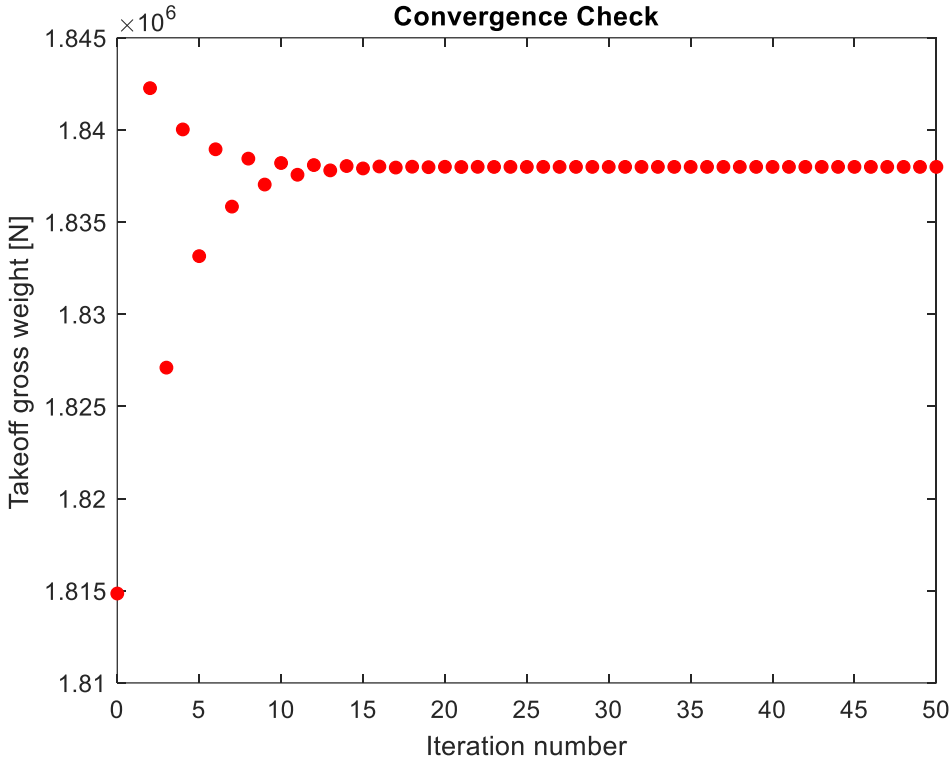
## 5. Example supersonic aircraft sizing code outputs: Concorde

Model status: These scripts are conceptual/preliminary engineering models. Several aerodynamic, propulsion, structural, and thermal relationships are analytical or empirical approximations. Results should not be interpreted as certification-level aircraft predictions. The purpose of the models is to investigate multidisciplinary aircraft-design relationships.

- TOGW=1726454.090574
- W_e=752692.709073
- Range=6396702.869683
- Lift to drag ratio=6.939458
- Drag_per_engine=46385.017084
- S=338.509790
- Alt=15800.000000
- V=590.136037
- V_a=220.999943
- c_r=23.653327
- c_t=3.547999
- b=24.889213
- Sweep=55.000000
- CPR=15.500000
- BR=0.000000
- FPR=0.000000
- A_inlet=0.951000
- Thrust_required=46385.017084
- TIT=1400.000000
- TSFC=0.000033
- S_error=5.510177
- b_error=2.624363
- c_r_error=14.485443
- TOGW_error=-4.870701
- We_error=-2.506880
- Wf_error=7.575761
- alt_error=1.250000
- R_error=-3.172627
- L/D_error=2.808705
- TSFC_error=1.437840


Design Point
×10^6
1.84
1.82
1.8
1.78
1.76
1.74
1.72
TOGW [N]
1.4
1.45
1.5
1.55
1.6
1.65
1.7
1.75
1.8
Altitude [m]
×10^4


Constraint Analysis
stall
land
takeoff
cruise
climb_1
turn
combined
end of climb
start of climb
3.5
3
2.5
2
1.5
1
0.5
0
Thrust to weight ratio (T/W) [-]
0
1000
2000
3000
4000
5000
6000
Wing loading (W/S) [N/m^2]


Convergence Check
×10^6
1.845
1.84
1.835
1.83
1.825
1.82
1.815
1.81
Takeoff gross weight [N]
0
5
10
15
20
25
30
35
40
45
50
Iteration number

# 6. Methods used in the sizing code in order:

## 1. Aerodynamics methodology to obtain the lift to drag ratio

- Obtain subsonic airfoil drag polar data of various airfoils using airfoiltools.com
- Begin the altitude sweep and TOGW convergence for-loops
- Calculate ambient atmospheric properties
- Use a simplified minimum pressure coefficient formula which feeds into the wing sweep solver. The wing sweep solver uses the isentropic pressure-ratio relation to solve for the critical Mach number. This and the Mach number determines the angle of the sweep.
- Solve for average weight during cruise and use it to size the aircraft.
- Calculate wing aspect ratio, wing area, and span, Oswald efficiency factor, root chord length, sectional and overall lift curve slopes. The wing area is calculated from the max takeoff gross weight and the sea level stall speed and air density.
- Calculate engine drag using the frontal area of each engine.
- Calculate the compression lift using the shock properties generated at the nose of the aircraft.
- Solve for the fuselage length (if desired) using the fuel volume as the fuselage volume minus 10 meters of length for non-fuel volume.
- Solve for the coefficient of skin friction using empirical relations
- Solve for the fuselage and wing parasite drag coefficient, zero-lift wave wing drag coefficient, and fuselage wave drag coefficients. Together, these make up the total zero-lift drag coefficient. The zero-lift wave uses a simplifying relation that depends on sweep and total airfoil thickness.
- Solve for the coefficient of lift using the mean weight and compression lift (if activated). For the subsonic case, the airfoil drag polar data is used to obtain the coefficient of lift and drag (adding induced drag in separately later).
- Solve for angle of attacks using the lift-curve slope and coefficient of lift.
- Solve for the coefficient of wave drag due to lift using a simplifying expression which uses the angle of attack (supersonic case).
- Solve for the induced drag coefficient (which I believe goes to 0 for the XB-70 if it applies wing droop).
- Solve for the total drag. For the supersonic case, the total drag is zero lift + induced + wave due to lift + engine drag. For the subsonic case, since drag polar data is used, The total drag is the drag polar drag + induced drag + fuselage parasite drag.
- Calculate the thermal protection system weight (supersonic) that depends on the weight of the insulation type material and the weight of the cooling jacket.
- Finally, calculate the lift to drag ratio that is fed into the Breguet fuel mass formula for the convergence of the TOGW at each altitude.

## 2. Propulsion methodology to obtain the thrust specific fuel consumption (TSFC)

- An if statement is created that controls which chunk of equations to solve: for the ideal engine or for an existing engine. For the ideal engine, a series of for-loops are created that solves for engine variables like bypass ratio, compressor pressure ratio, and turbine inlet temperature that must produce a thrust equal to the drag of the aircraft per engine.
- The inlet, compressor, fan, and turbine total temperatures and pressures are solved for.
- The fuel flowrate per oxygen flowrate fraction is solved for using the turbine inlet temperature.
- The exit temperature is solved for and is used to calculate the engine core and fan exit velocity.
- A for loop is created to solve for the sea level mass flow rate. The kinetic energy formula (Ke_dot=.5*m_dot*Ve^2) is used to help solve for m_dot.
- Next, the core thrust is solved for and adjusts the turbine inlet temperature to thrust match the drag of the aircraft. If the total turbine inlet temperature becomes maxed out, the afterburner is applied for the remaining thrust.
- The afterburner exit velocity is solved for by using the specified total afterburner temperature which solves for the fuel fraction of the afterburner required to generate the remaining thrust that the core itself cannot generate without the afterburner.
- Next, the thrust specific fuel consumption is solved for.
- Next, the engine weight is solved for (if solving for ideal engine) using Raymer's approximations that depend on the sea level thrust.

## 3. Structural methodology to obtain weights

- Raymer's approximations are used to determine all the different weight components of the empty weight.
- Next, the empty weight is found by adding the listed weights together plus the skin, thermal protection system (supersonic), and payload weights.

## 4. Performance and trajectory methodology for climb and descent

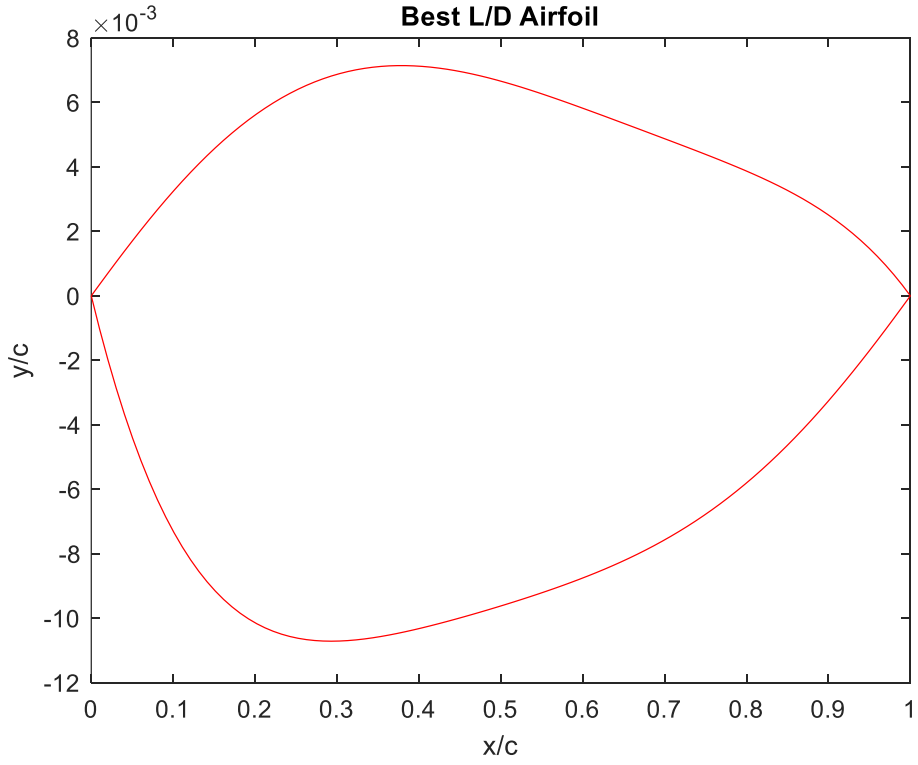
- The total fuel required for the descent weight is found first by starting with the landing weight and sink rate.
- Next, the Breguet fuel mass is calculated using this weight at the end of cruise to determine the weight at the beginning of cruise.
- Next, the total fuel required for the climb phase is calculated by starting with the initial cruise weight and climb rate.
- Finally, the total takeoff gross weight is found by adding the climb weight to the initial cruise weight. This value is the TOGW used for each altitude.

## 5. Carpet plots (constraint diagrams)

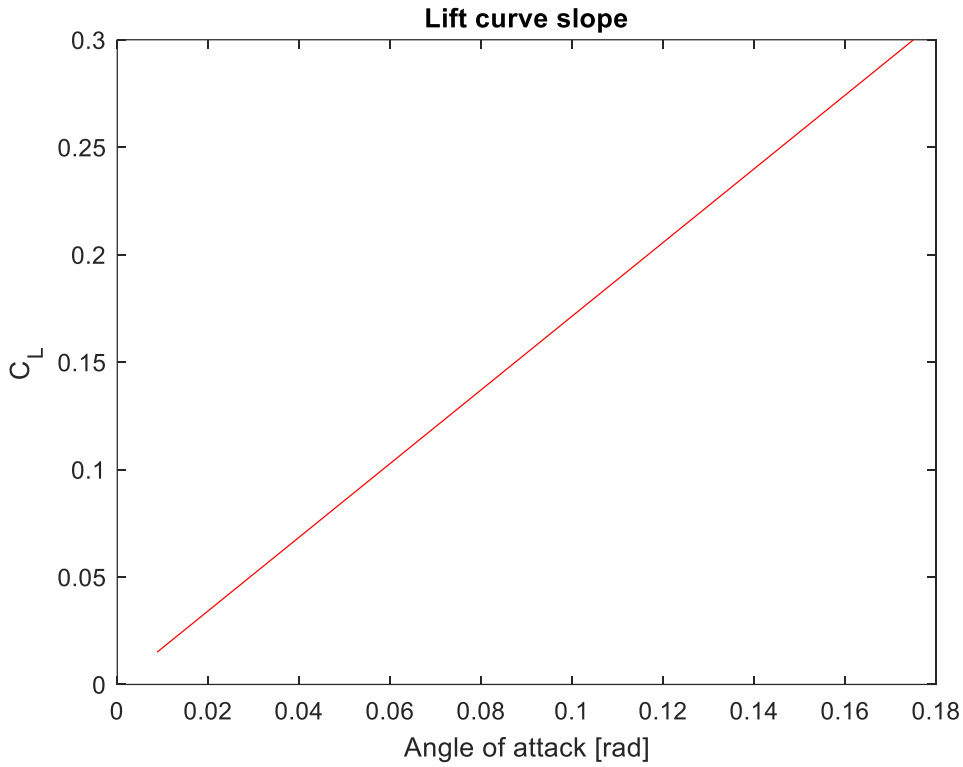
- The wing loading equations for stall and land are specified.
- The thrust to weight ratio for takeoff, climb, cruise, and turn are specified. These equations use the required thrust. The takeoff distance includes clearing a 50ft obstacle.
- Next, the initial climb and initial cruise design point is plotted against these curves. These two points use the available thrust.
- Next, the upper left space defined by the curves is the area for feasibility.

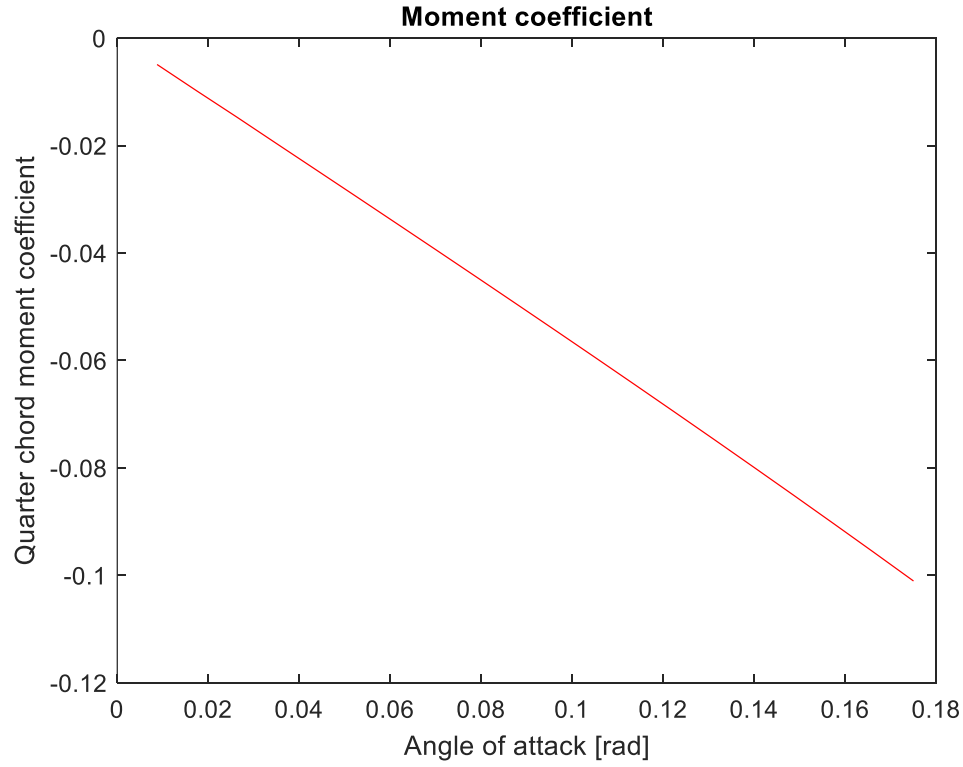
## 7. Airfoil optimization code outputs for the Concorde flight condition

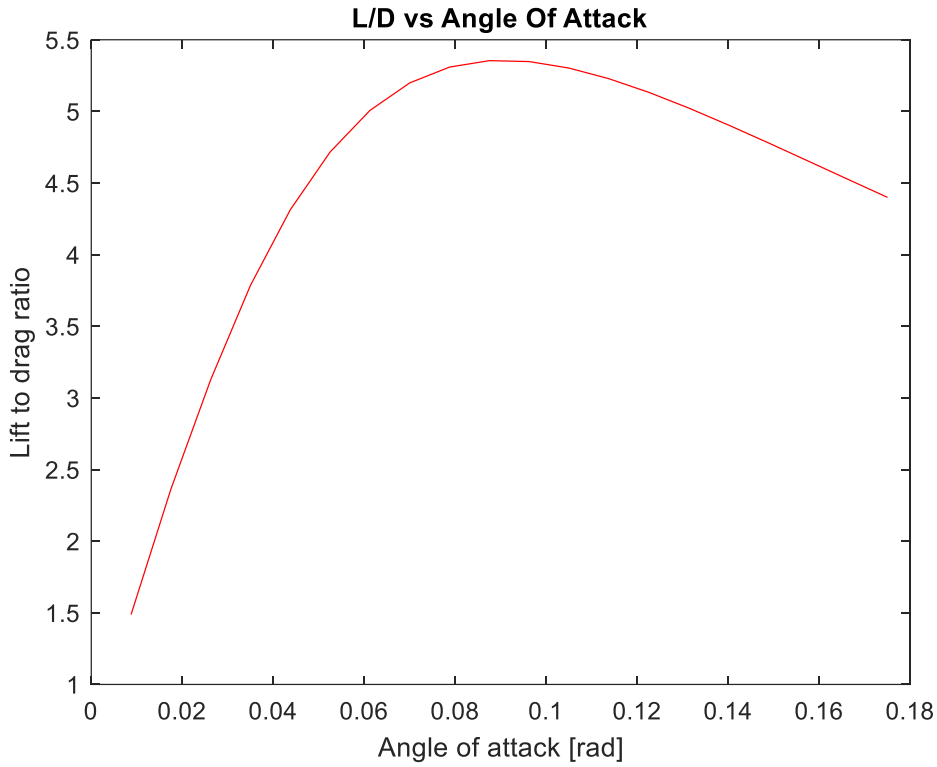
- aoa=0.063866
- y_crest_u=0.007143
- y_crest_l=-0.010714
- x_crest_u=0.378571
- x_crest_l=0.292857
- L/D=5.074093

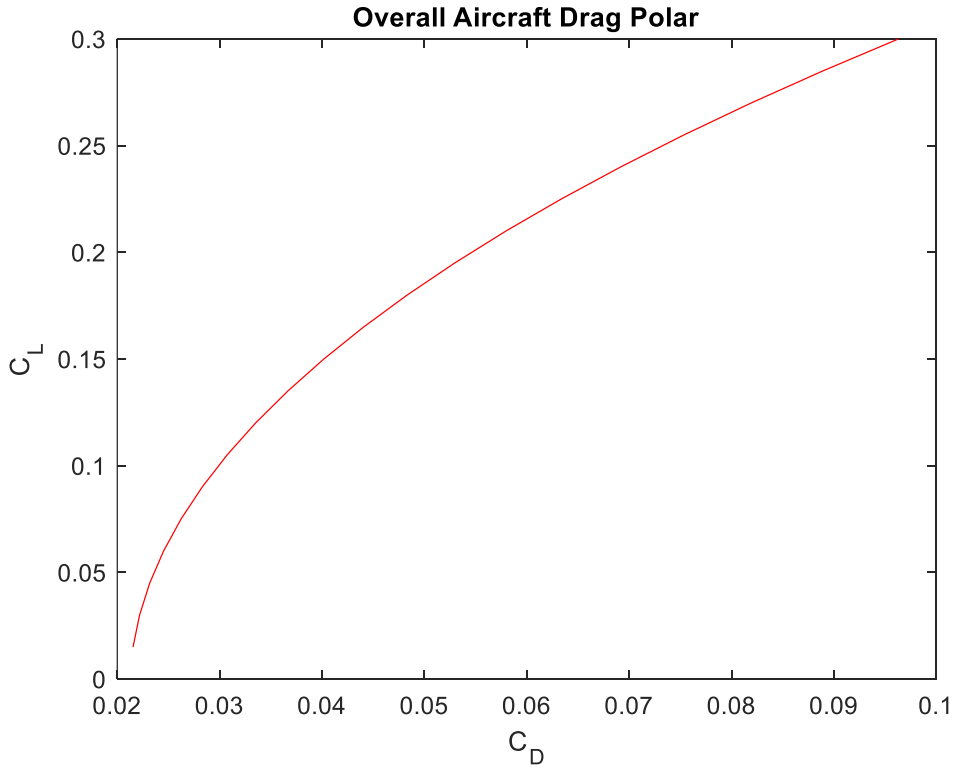


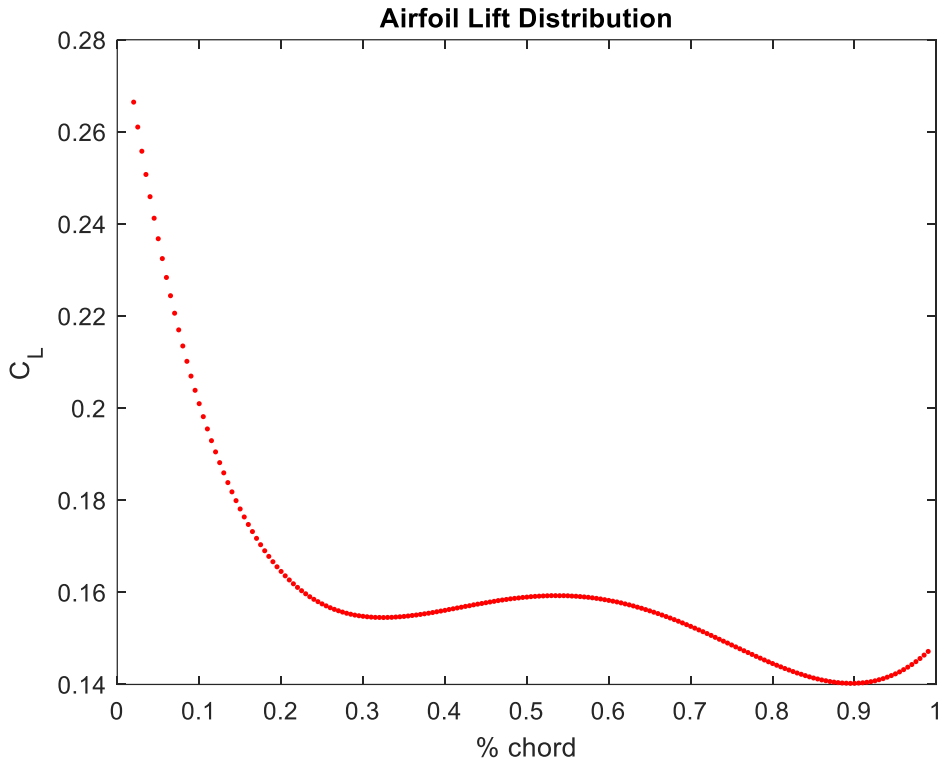
.


Lift curve slope
0.3
0.25
0.2
0.15
0.1
0.05
0
$C_L$
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
Angle of attack [rad]


Moment coefficient
0
-0.02
-0.04
-0.06
-0.08
-0.1
-0.12
Quarter chord moment coefficient
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
Angle of attack [rad]


L/D vs Angle Of Attack
Lift to drag ratio
Angle of attack [rad]
5.5
5
4.5
4
3.5
3
2.5
2
1.5
1
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18


Overall Aircraft Drag Polar
C_L
C_D
0.3
0.25
0.2
0.15
0.1
0.05
0
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1


Airfoil Lift Distribution
0.28
0.26
0.24
0.22
0.2
0.18
0.16
0.14
$C_L$
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
% chord

## 8. Methods used in the airfoil optimization code in order:

- For loops of the max thickness location and max thickness of the upper surface and lower surface air foil are created to iterate for the best combination.
- Next, the coefficients of a polynomial are solved for using the geometric input variables of the airfoil. The subsonic polynomial takes the form y/c = a1*x^(1/2) + a2*x + a3*x^2 + a4*x^3 + a5*x^4. The supersonic polynomial takes the form y/c = a1*x + a2*x^2 + a3*x^3 + a4*x^4 + a5*x^5.
- Next, for the subsonic case, for-loops are created to solve for the A coefficients and zero lift angle of attack of thin airfoil theory. For the supersonic case, shock and expansion relations are used.
- Next, the velocity at each point along the chord is calculated using the A coefficients for the upper and lower surface for the subsonic case.
- Isentropic relations are then used to calculate the pressure coefficient at each point along the chord.
- Next, the axial and normal force coefficients are solved for and converted into the coefficient of lift and drag with the ratio calculated.
- Next, for the subsonic case, the equations for Stratford's flow separation criteria are solved for:
    - One thing thin airfoil theory does not take into account 1) The minimum pressure coefficient that forms at the leading-edge radius of the wing for the upper surface dictated by the leading-edge suction-pressure coefficient formula. 2) The stagnation pressure where the angle of the upper surface is high compared to the angle of the freestream angle.
    - Next, the stagnation effects are added to the thin airfoil theory coefficient of pressure that is used to obtain the minimum pressure coefficient along the upper surface. This is used in the equations for Stratford's separation criteria.
    - The canonical pressure coefficient is then solved for that feeds into Stratford's separation equations.
- Next, the lift to drag ratio is compared for each combination of geometric variables of the airfoil. The highest lift to drag ratio is the output airfoil.

## 9. Sizing code Inputs: Concorde

```
% ==================== MISSION REQUIREMENTS
====================
M = 2; % Cruise Mach number [-]
Range_cruise = 5600000; % Cruise range [m]
TOGW_i(1) = 185000*9.81; %Initial takeoff gross weight
W_e(1)    = 78700*9.81; %Initial empty weight
W_payload = 13400*9.81; % payload weight [N]
W_f(1)    = TOGW_i(1)-W_e(1)-W_payload; % Initial fuel
weight [N]
Np        = 100; % Passenger capacity [-]
type2     = 0; % Aircraft family: 0=transport, 1=GA,
2=fighter/delta family


% ==================== PROPULSION ====================
type        = 0; % Propulsion type: 0 = existing engine, 1
= ideal engine
n           = 4; % Number of engines used in propulsion
calculation [-]
N_en        = 4; % Total number of engines [-]
Afterburner = 1; % Afterburner/reheat: 0=disabled, 1 =
enabled
Scramjet    = 0; % Scramjet: 0 = disabled, 1 = scramjet, 2
= ramjet
Waverider   = 0; % Waverider configuration: 0 = disabled, 1
= enabled
A_inlet     = .951; % Inlet capture area per engine [m^2]
TIT_d_low   = 1200; % Low design turbine inlet temperature
[K]
TIT_d_high  = 1400; % High design turbine inlet temperature
[K]
T_0_6_high  = 1800; % High turbine/combustor total
temperature [K]
T_0_3_sl    = 550; % Sea-level compressor total temperature
which the sea level thrust depends on [K]
CPR_min     = 15.5; % Minimum compressor pressure ratio [-]
CPR_max     = 15.5; % Maximum compressor pressure ratio [-]
BR_min      = 0; % Minimum bypass ratio [-]
BR_max      = 0; % Maximum bypass ratio [-]
FPR         = 0; % Fan pressure ratio [-]
n_ab_low    = 0; % Low afterburner setting [-]
n_ab_high   = 0.6; % High afterburner setting [-]
```

```
eta_inlet   = 0.92; % Inlet efficiency at M=2 [-]
eta_c       = 0.85; % Compressor efficiency [-]
eta_f       = 0.88; % Fan efficiency [-]
eta_t       = 0.88; % Turbine efficiency [-]
eta_b       = 0.98; % Burner/combustor efficiency [-]
eta_ab      = 0.85; % Afterburner efficiency [-]
eta_m       = 0.98; % Mechanical efficiency [-]
eta_n       = 0.94; % Nozzle efficiency [-]
Cd_engine   = 0.278; % Engine/nacelle drag coefficient [-]
rho_f       = 800; % Fuel density [kg/m^3]

% ==================== EXISTING ENGINE DATA (Olympus 593)
====================
Th_wet          = 169200; % Maximum wet/reheat thrust per
engine [N]
Th_wet_cruise   = 140000; % Cruise wet/reheat thrust per
engine [N]
Th_dry          = 142300; % Maximum dry thrust per engine
[N]
Th_dry_cruise   = 140000; % Cruise dry thrust per engine [N]
TIT_e_low       = 1200; % Engine low turbine inlet
temperature [K]
TIT_e           = 1400; % Engine turbine inlet temperature
[K]
CPR_e           = 15.5; % Engine compressor pressure ratio
[-]
BR_e            = 0; % Engine bypass ratio [-]
FPR_e           = 0; % Engine fan pressure ratio [-]
eta_t_e         = 0.88; % Engine turbine efficiency [-]
eta_f_e         = 0.88; % Engine fan efficiency [-]
eta_ab_e        = 0.85; % Engine afterburner efficiency [-]
eta_c_e         = 0.85; % Engine compressor efficiency [-]
eta_m_e         = 0.98; % Engine mechanical efficiency [-]
eta_b_e=.98;                    % Engine burner efficiency
[-]
A_inlet_e       = .951;1.125; % Existing engine inlet area
[m^2]
T_0_6_k_e       = 1800; % Existing engine turbine
temperature [K]
W_eng_e         = 3175*9.81; % Engine weight per engine [N]
(~7000 lbf class)

% ==================== AERODYNAMICS ====================
Subsonic_airfoil = 5; % Airfoil model/index [-]
```

```
delta        = 0; % Delta-wing flag: 0 =
conventional/defined geometry
sweep_solver = 0; % Sweep calculation method/index [-]
x_c          = 0.5; % Airfoil maximum thickness/camber
reference location [-]
cl_max       = 1.35; % 2-D airfoil maximum lift coefficient
[-]
CL_max       = 1.3; % Aircraft maximum lift coefficient [-]
(low-AR delta)
tdc          = 0.03; % Maximum thickness-to-chord ratio [-]
V_stall_sl   = 80; % Sea-level stall speed [m/s]
c_f          = 0.0020; % Skin-friction coefficient [-]
u_0          = 1.7894e-5; % Reference dynamic viscosity
[Pa*s]
Su           = 110.4; % Sutherland constant [K]
nose_angle   = 5; % Nose half angle
sweep        = 55*pi/180; % Representative LE/ogive sweep
[rad]
sweep_q      = 55*pi/180; % Quarter-chord/analysis sweep
[rad]
A_base       = 8; % Base area [m^2]
lambda       = 0.15; % Taper/geometry parameter [-]
droop        = 0; % Drooped nose: 0 = no droop
kA           = 0.2; % Wing volume coefficient [-]
AR           = 1.83; % Wing aspect ratio [-]

%% ================ CONCORDE TAIL INPUTS ================

% Horizontal tail
S_H       = 0; % Horizontal-tail planform area [m^2]
b_H       = 0; % Horizontal-tail span [m]
sweep_H   = 0; % Horizontal-tail sweep [rad]
AR_H      = 0; % Horizontal-tail aspect ratio [-]
e_H       = 0; % Horizontal-tail Oswald efficiency [-]
CL_H      = 0; % Horizontal-tail lift coefficient [-]
alpha_H   = 0; % Horizontal-tail angle of attack [rad]
c_H  = 5.0; % Horizontal-tail mean aerodynamic chord [m]
x_t_H = 0.30; % Horizontal-tail maximum thickness location,
x/c [-]
t_c_H = 0.08; % Horizontal-tail thickness-to-chord ratio [-
]


% Vertical tail / fin
```

```
S_V       = 33.91; % Vertical-tail/fin planform area [m^2]
b_V       = 11.32; % Vertical-fin height/span [m]
sweep_V   = 51*pi/180; % Vertical-fin leading-edge sweep
[rad]
AR_V      = b_V^2/S_V; % Vertical-fin aspect ratio [-]

e_V       = 0.80; % Vertical-fin Oswald efficiency [-]
CL_V      = 0; % Vertical-fin lift coefficient [-]
beta_V    = 0; % Sideslip angle [rad]

t_c_V     = 0.046; % Vertical-tail maximum thickness-to-
chord ratio [-]

c_V  = 5.5; % Vertical-tail mean aerodynamic chord [m]
x_t_V = 0.30; % Vertical-tail maximum thickness location,
x/c [-]
t_c_V = 0.08; % Vertical-tail maximum thickness-to-chord
ratio [-]

% ==================== ATMOSPHERE / EARTH
====================
Cp = 1009; rho_sl = 1.226; T_sl = 288.15; R = 287.05; %
Specific heat of air [J/(kg*K)]
y = 1.4; g = 9.81; p_sl = 101300; radius_e = 6.371e6; %
Ratio of specific heats [-]
mu_0 = 1.716e-5; % Reference dynamic viscosity [Pa*s]
T_0  = 273.15; % Reference temperature [K]
Su    = 110.4; % Sutherland constant [K]

% ==================== STRUCTURAL / RAYMER GEOMETRY
(Concorde) ====================
N_z = 3.75; % Ultimate load factor / structural factor [-]
N_l = 3.0; % Landing load factor / structural factor [-]
ratio_fin = 19; % Fin geometry ratio [-]
D_f    = 3.10; % Fuselage maximum diameter [m]
L      = 61.66; % Fuselage/aircraft length (0 = solve for)
[m]
W_fuse = 2.88; % Fuselage width [m]
p_cabin = p_sl; % Cabin pressure [Pa]
t_aluminum = 0.002; % Aluminum skin thickness [m]
rho_aluminum = 2800; % Aluminum density [kg/m^3]

S      = 358.25; % Wing reference area [m^2]
S_csw  = 28.0; % Elevon/control-surface area [m^2]
```

```
S_ht   = 0.1; % Horizontal-tail/canard area [m^2]
S_vt   = 32.0; % Vertical-tail area [m^2]
B_h    = 6.0; % Horizontal-tail span/reference dimension
[m]
F_w    = 2.88; % Fuselage width [m]
H_t_Hv = 0; % Horizontal-tail to vertical-tail parameter [-
]
L_t    = 22.0; % Tail moment arm [m]
Sr_Svt = 0.35; % Rudder/control surface to vertical-tail
area ratio [-]
A_vt   = 1.4; % Vertical-tail geometry parameter [-]
sweep_vt = 50*pi/180; % Vertical-tail sweep [rad]
S_cs   = S_csw + 0.35*S_vt; % Total control-surface area
[m^2]

L_m  = 3.50; % Main landing gear length [m]
L_n  = 2.80; % Nose landing gear length [m]
N_mw = 8; % Main landing gear wheels [-]
N_nw = 2; % Nose landing gear wheels [-]
N_mss = 2; % Main landing gear struts/assemblies [-]
L_s  = 3.50*3.28; % Landing gear strut length [ft]

K_dw = 0.768; K_vs = 1.0; K_dwf = 0.774; % Design wing
weight correction factor [-]
K_rht = 1.0; K_uht = 1.0; K_door = 1.06; K_Lg = 1.12; %
Horizontal-tail correction factor [-]
K_cb = 1.0; K_tpg = 1.0; K_mp = 1.0; K_np = 1.0; % Cabin
correction factor [-]
K_vg = 1.0; K_d = 1.75; K_ws = 0.0; Ls_Ld = 0.40; %
Variable-geometry correction factor [-]
K_sys=1.65; % Systems weight constant [-]
K_gr = 0.007; % Landing gear weight coefficient [-]
a_g  = 0.95; % Landing gear weight exponent [-]
b_g  = 0.60; % Landing gear strut length exponent [-]


D_e     = 1.21; % Equivalent fuselage/cabin diameter [m]
L_d     = 8.0; % Door/fuselage length parameter [m]
L_tp    = 2.0; % Tail-pipe length [m]
L_sh    = 1.5; % Shaft/structural length parameter [m]
L_ec    = 25.0; % Equipment/cabin length [m]

V_t   = 119.5; % Total fuel tank volume [m^3]
Vi_Vt = 0.90; % Usable/installed tank volume ratio [-]
```

```
Vp_Vt = 0.10; % Tank volume fraction [-]
N_t   = 17; % Number of fuel tanks [-]
SFC   = 1.19; % Cruise specific fuel consumption
[lb/(lbf*hr)]

N_c = 3; N_ci = 2.0; N_s = 2; N_u = 8; % Number of crew [-]
Nf = 5; Nm = 2; W_uav = 800*9.81; % Number of
furnishings/functional units [-]

K_vsh = 1.0; % Vertical/horizontal stabilizer systems
factor [-]
K_mc  = 1.0; % Mechanical/control systems factor [-]
R_kva = 80; % Electrical system rating [kVA]

N_doors = 4; N_cargo_doors = 1; % Number of
passenger/service doors [-]
S_door = 1.5; S_cargo_door = 2.0; % Passenger/service door
area [m^2]
S_floor_main = 80; S_floor_upper = 0; % Main-deck floor
area [m^2]
w_main = 2.1/0.0208854; w_upper = 0; % Main-deck floor
areal weight [lb/ft^2]

%% ---- nacelle / intake (4 integrated 2-D ramps) ----
K_ng     = 1.0; % Nacelle installation factor: not pylon-
mounted [-]
K_p      = 1.0; % Pylon factor [-]
K_tr     = 1.18; % Thrust reverser factor [-]
N_Lt     = 8.0; % Intake + engine characteristic length [m]
N_w      = 1.40; % Equivalent nacelle/inlet width [m]
S_n      = pi*N_w*N_Lt; % Equivalent nacelle
wetted/reference area [m^2]
N_e      = 4; % Number of engines [-]
W_en     = W_eng_e; % Engine weight per engine [N]
N_Lt_ft  = N_Lt/0.3048; % Intake + engine length [ft]
N_w_ft   = N_w/0.3048; % Equivalent nacelle width [ft]
S_n_ft   = S_n/0.3048^2; % Equivalent nacelle area [ft^2]
W_en_lb  = W_eng_e/4.44822; % Engine weight per engine [lb]

% ==================== PERFORMANCE / TRAJECTORY
====================
rate_of_climb    = 5000/3.28/60; % Rate of climb [m/s]
rate_of_descent  = 3000/3.28/60; % Rate of descent [m/s]
Takeoff_field_length = 3400; % Takeoff field length [m]
```

```
Landing_field_length = 2200; % Landing field length [m]
g_turn = 1.2; % Turning/acceleration load factor [-]
mu_2 = 0.03; % Ground friction coefficient [-]
alt_0 = 1000; % Initial altitude [m]
acc = 2.0; % Acceleration [m/s^2]
V_lof = 1.1*V_stall_sl; % Lift-off speed [m/s]

% ==================== AEROTHERMAL PARAMETERS
====================
TPS=0; % Thermal protection system: 0=no, 1=yes
R_wle = 0.344/12; % Wing leading-edge radius [ft]
g_s=g*3.2797; % Gravity conversion parameter [ft/s^2]
J=778; % Mechanical equivalent of heat [ft*lbf/BTU]
emmissivity=0.8; % Radiative emissivity [-]
Cp_wall=1000; % Wall/TPS specific heat [J/(kg*K)]
rho_TPS=40; % TPS areal density [kg/m^2]
rho_titanium=4510; % Titanium density [kg/m^3]
T_ins_max=581; % Maximum insulation temperature [K]
T_sys_max=273+70; % Maximum systems temperature [K]
k_titanium=21; % Titanium thermal conductivity [W/(m*K)]
k_insulation=.3; % Insulation thermal conductivity
[W/(m*K)]
T_initial=230; % Initial internal air temperature [K]
t_ins=.051; % Insulation thickness [m]
t_tit=.001;.0015; % Titanium thickness [m]
Cp_tit=523.5; % Titanium specific heat [J/(kg*K)]
Cp_ins=500; % Insulation specific heat [J/(kg*K)]
rho_ins=112;270; % Insulation density [kg/m^3]
S_cooling=39;20;39; % Active cooling surface area [m^2]
rho_cooling=300; % Active cooling surface areal density
[kg/m^2]
x = 4; % Heating-analysis longitudinal location [m]
Pr = 0.72; % Prandtl number [-]

% ==================== FUEL PROPERTIES ====================
E_sp=44E6; % Fuel specific energy [J/kg]
ratio_fo=.294; % Stoichiometric fuel-to-oxygen mass ratio
[-]
Cp_f=2010; % Fuel specific heat [J/(kg*K)]

% ==================== ITERATION / SEARCH PARAMETERS
====================
min_alt=14000; % Minimum search altitude [m]
max_alt=18000; % Maximum search altitude [m]
```

```
WdS_max=6000; % Maximum wing loading shown in carpet plot
[N/m^2]
alt_points=200; % Number of altitude points shown in design
point plot [-]
i_max=20; % number of iterations for finding the ideal
engine compressor pressure ratio during cruise [-]
ii_max=25; % number of iterations for finding the existing
engine turbine inlet temperature during cruise [-]
j_max=20; % number of iterations for finding the ideal
engine bypass ratio during cruise [-]
k_max=20; % number of iterations for finding the ideal
engine turbine inlet temperature during cruise [-]
q_max=50; % number of iterations for the takeoff gross
weight using the Breuget fuel weight for each iteration
dd_max=15; % number of iterations from altitude=0 to
altitude=design for the descent phase
cc_max=15; % number of iterations from altitude=0 to
altitude=design for the climb phase
zzz_max=30; % number of points used for the carpet plot
that has smoothed out curves
xx_max=20; % number of iterations for finding the shockwave
angle generated at the nose of a supersonic aircraft
iii_max=30; % number of iterations for the stagnation
temperature during aerodynamic heating of a high speed
supersonic aircraft
ff_max=30; % number of iterations for finding the existing
engine afterburner fuel fraction flowrate
kk_max=30; % number of iterations for finding the ideal
engine afterburner fuel fraction flowrate
kk_Dmax_max=30; % number of iterations for finding the
ideal engine afterburner fuel fraction flowrate at the max
drag condition
f_6_k_e_i_low=0; % Lower bound for existing engine
afterburner fuel fraction flow rate
f_6_k_e_i_high=.5; % Upper bound for existing engine
afterburner fuel fraction flow rate


% --- Concorde ---

S0      = 358.25; % Reference wing area [m^2]
b0      = 25.56; % Reference wingspan [m]
c_r0    = 27.66; % Reference root chord [m]
Lambda0 = 0; % Reference wing sweep [rad]
```

```
Alt0    = 16000; % Reference cruise altitude [m]
R0      = 6200e3; % Reference range [m]
L_D     = 7.14; % Reference cruise lift-to-drag ratio [-]
TSFC0   = 33.8e-6; % Reference cruise TSFC [kg/(N*s)]
```

# 10. Airfoil optimization code inputs: Concorde

```
C_L=.1*.7;                % 3-D lift coefficient target
sweep=65.5*pi/180;        % wing sweep [deg then rad]; last value used
e=.4;                     % Oswald efficiency
AR=1.75;                  % aspect ratio
C_d_0=.0075;              % profile / parasite Cd
Cd_p_fus=.008;            % fuselage parasite Cd
i_max=200;                % chordwise stations for force integration
xx_max=200;               % stations for thin-airfoil integrals
y=1.4;                    % gamma (ratio of specific heats)
M=3;                      % Freestream Mach; first value used
a_inf=343;                % Freestream air speed of sound
mu=1.789E-5;              % Freestream air dynamic viscosity
rho=.3;                   % Freestream air density
T_inf=300;                % Freestream air temperature
p_inf=.2*101300;          % Freestream air pressure
Cp=1005;                  % Specific heat of air
c_root=11.42;             % Root chord length
%% ======================== 2. AIRFOIL GEOMETRY CONTROLS
========================
r_le = 0.04;              % Wing leading-edge radius / chord
theta_le_u = 2*pi/180;    % Upper surface leading edge angle
theta_te_u = -2*pi/180;   % Upper surface trailing edge angle
theta_le_l = -6*pi/180;   % Lower surface leading edge angle
theta_te_l = 2*pi/180;    % Lower surface trailing edge angle

x_crest_u_min=0.25;       %Lower bound best L/D max airfoil thickness x location (upper
surface)
x_crest_u_max=0.4;        %Upper bound best L/D max airfoil thickness x location (upper
surface)
y_crest_u_min=0.005;      %Lower bound best L/D max airfoil thickness y location (upper
surface)
y_crest_u_max=.01;        %Upper bound best L/D max airfoil thickness y location (upper
surface)
x_crest_l_min=0.25;       %Lower bound best L/D max airfoil thickness x location (lower
surface)
x_crest_l_max=0.4;        %Upper bound best L/D max airfoil thickness x location (lower
surface)
y_crest_l_min=-0.005;     %Lower bound best L/D max airfoil thickness y location (lower
surface)
y_crest_l_max=-.015;      %Upper bound best L/D max airfoil thickness y location (lower
surface)

%% ============================ For loop iteration parameters
======================== %%
N=20;                     % bisection iterations for Prandtl-Meyer / shock
xxxx_max=15;
```

```
xxxxx_max=20;
xxxxxxx_max=15;
pp_max=7;
nn_max=7;
p_max=7;
n_max=7;
```